Effects of JAK-STAT pathway on melanotic tumors induced by juvenile hormone mimics

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Abstract

It is recently found that juvenile hormone mimics (JHMs) such as methoprene, which are widely used as insecticides, induce melanotic tumors composed of lamellocytes. However, the specific immune pathway utilized by lamellocyte proliferation is not yet known. This experiment tests whether JAK-STAT is the immune pathway used by knocking down the Domeless receptor using RNA interference. We hypothesized that because JAK-STAT is known to mediate lamellocyte production, knocking down Domeless – the sole receptor of the JAK-STAT pathway – would decrease JHM-induced tumor formation. Using the Cg-GAL4 driver to knock down Dome, Ebony, and White genes and treating pupae with either ethanol (control) or methoprene (JHM), we measured tumor formation and viability. Methoprene caused tumors to form in all three RNAi groups, and the chi-square test revealed no significant difference between the Dome RNAi and White RNAi groups. This suggests that JAK-STAT is not the immune pathway that is utilized, which does not support our hypothesis, and that JHM-induced melanotic tumor formation is mediated by an immune pathway other than JAK-STAT, such as the Toll pathway.

Introduction

Juvenile hormone (JH) is a sesquiterpenoid and plays a critical role in the nymphal and larval developmental stages of all insects (Riddiford, 2009). It maintains the juvenile state by repressing the switching actions of ecdysone, such as pupariation and metamorphosis, and ensures that the fly can progress through its three instar phases (James, 2011). Essentially, the JH moderates the effects of ecdysone during each molt, preventing premature metamorphosis. JH is produced by the corpora allata, a paired gland in the fly's head. Normally, endogenous JH levels decline at the end of the third and final instar phase, allowing ecdysone to induce metamorphosis. When JH or juvenile hormone mimics (JHMs) are applied at the end of this instar phase, D. melanogaster larvae exhibit a lengthened final instar phase (Riddiford & Ashburner, 1991). In some cases, high doses of JH can even lead to death during the pupal-adult transition. Thus, because JH mediates many developmental pathways and can even lead to lethality, juvenile hormone mimics (JHMs), like methoprene, are widely used as insecticides (Bailly & Kohlmeier, 2025). Unlike conventional insecticides that solely induce mortality, methoprene, which is chemically and functionally similar to natural JH, impedes the fly's ability to mature into reproductive adults. Methoprene is used for many purposes, such as controlling mosquito larvae to prevent the spread of diseases (e.g. malaria, Zika virus) or protecting crops from pests. In addition to developmental effects, methoprene and pyriproxyfen, another JHM, have been seen to cause melanotic tumor proliferation composed of lamellocytes (Luu et al., 2025). It has been proven that the juvenile hormone receptors Methoprene-tolerant (Met) and germ-cell expressed (gce) mediate this reaction, but it is still unclear what specific pathways are downstream of this process. Investigating these downstream pathways requires precise control over gene expression in immune tissues, which can be achieved through the GAL4-UAS system.

The GAL4-UAS is a two-part tool used in Drosophila to control gene expression in tissue-specific cells. It consists of the GAL4 transcription factor, placed on the driver line, and the UAS enhancer, placed on the responder line. The protein GAL4, which is derived from yeast, is a transcriptional activator, meaning it binds to specific DNA sequences to increase the transcription of a gene (Brand & Perrimon 1993). The gene that encodes for GAL4 is placed under the control of a tissue-specific promoter or enhancer, causing the protein to only be produced in specific tissues. This allows the scientists to manipulate a gene in only one tissue and study tissue-specific function without killing the whole organism. UAS, the other component, is an enhancer that is placed upstream of the gene of interest and thus controls the expression of that gene. The transcription of UAS requires the presence of GAL4 as GAL4 binds to UAS: without GAL4, UAS is not transcribed, and the gene of interest is not expressed; with GAL4, UAS is transcribed and the gene of interest is expressed. (Duffy, 2002) In order to activate their transcription, flies with the responder line containing UAS and flies with the driver line containing GAL4 are crossed, producing progeny that has GAL4 to activate the transcription of the gene of interest. Cg-GAL4 is a specific driver used in Drosophila research used to target genes in the fly's fat body (metabolic/liver tissue) and their hemocytes (Schmid et al., 2014). Cg

refers to the Collagen IV promoter, which is active when collagen binds to it. The fly's fat body and hemocytes are highly active in immune responses, which is why it is used, along with RNA interference, to manipulate the JAK/STAT immune pathway.

RNAi, also known as RNA interference, is a phenomenon in organisms where the introduction of RNA (often in small pieces) can shut down protein translation by attaching themselves to the mRNA that codes the proteins. This leads to the silencing of genes and them not being expressed, as shown by Fire and Mello's experiment with C. elegans roundworms. They injected dsRNA (double stranded RNA) that mimicked unc-22 in the organisms, resulting in the mimicked gene becoming silenced. They found that the silencing spread throughout the body and became hereditary. Not only that, but it was a universal phenomenon in all organisms. The RNAi mechanism works by dsRNA binding to the protein Dicer, which breaks the dsRNA up into small interfering RNAs (siRNAs) (Bernstein, 2001). During strand separation, only one of the strands – the guide/antisense strand – is retained and loaded into the RISC complex (Hammond, 2000), while the other passenger/sense strand is degraded (Khvorova & Schwarz, 2003). The guide strand then base-pairs with the target mRNA, allowing the RISC complex to destroy the mRNA and halt translation of the protein, thereby preventing gene expression. Domeless is the transmembrane receptor for the JAK/STAT signaling pathway. Upon activation by ligands of the Unpaired (Upd) family, Domeless activates the intracellular Janus kinase (JAK), which leads to the phosphorylation of STAT family members that regulate gene expression. Through this signal transduction, the Dome gene has been found to participate in several biological processes, including pattern specification, imaginal disc development, cell population proliferation, and compound eye morphogenesis (FlyBase). Because Dome is the only receptor in the JAK/STAT pathway, the loss of Dome results in a failure to transduce Upd ligand signals, demonstrating that Dome is essential for this single pathway (Brown, 2001). More specifically, in the lymph gland – which contains two cell types, prohemocytes and hemocytes – the posterior signaling center (PSC) regulates the differentiation between these two types by controlling the secretion of Upd. Therefore, as the only receptor in the JAK/STAT pathway, Dome maintains the signaling activity that controls the differentiation of prohemocytes (Brown, 2001).

It is still unclear what pathway mediates the formation of melanotic tumors due to JHMs. Our experiment investigates the JAK-STAT immune pathway, which has responses that range from proliferation and differentiation to apoptosis. (Harrison, 2012) Moreover, the JAK-STAT has also been seen to mediate lamellocyte formation, shown in an experiment using hopscotch^Tumorous-lethal that has constitutively active JAK-STAT signaling. (Harrison et al., 1995) We hypothesized that because the JAK/STAT pathway is known to mediate lamellocyte production, supported by the hop^Tum-l mutation, if Domeless is knocked down by RNAi, then the amount of melanotic tumors due to JHMs such as methoprene will decrease. Without the Domeless receptor, the JAK/STAT pathway will not be able to activate and thus the formation of lamellocytes, which make up the melanotic tumors, will cease.

Methods

Drosophila melanogaster stocks were obtained and prepared by Mr. Luu: one Cg-GAL4 driver line and three RNA interference lines. The Cg-GAL4 driver line activates GAL4 expression in the fat body and hemocytes. Three RNA interference lines controlled by UAS were used: Dome-RNAi, which knocks down the Domeless gene; Ebony-RNAi, which knocks down a gene involved in body pigmentation; and White-RNAi, which knocks down a gene involved in eye pigmentation. The ebony and white RNAi were used as controls as they are not involved in immune function nor melanotic tumor formation. In order to knockdown tissue-specific genes, Cg-GAL4 flies were crossed with all three RNAi flies. The progeny had the Cg-GAL4 promoter along with each UAS-controlled RNAi, activating the RNAi knockdown in the specific tissue. Each cross was divided into treatment and treated by Mr. Luu with either 25 μL of ethanol (control) or methoprene at a concentration of 1 μg/μL dissolved in ethanol. Ethanol, which was 200 proof, was used as a control, specifically a vehicle control, because the JHM methoprene was delivered in ethanol. By using ethanol as a control, the possibility that ethanol caused the formation of melanotic tumors, rather than the JHM, is eliminated. In total, there were six experimental groups with varying sample sizes: Dome RNAi control treated with ethanol (n=4), Dome RNAi treated with methoprene (n=10), Ebony RNAi control treated with ethanol (n=25), Ebony RNAi treated with methoprene (n=25), White RNAi control treated with ethanol (n=25), and White RNAi treated with methoprene (n=25).

Larvae were scored at the wandering third instar and prepupal stages. Each vial was scored, and larvae were classified as either having melanotic tumors (melanized masses) or no tumors. Empty pupal cases were also recorded. Viability was also assessed: each vial was scored at the late pupa stage and classified as either full pupae (containing a dead adult that failed to eclose) or empty pupal cases (successful eclosion).

Vials with incomplete tumor counts or pupae viability counts were excluded from the analysis. The sample sizes in the methods section reflects the amount of vials before the exclusion, while the sample sizes in the results reflects the amount after the exclusion. Therefore, the sample sizes listed in those two sections may be inconsistent with each other.
For each vial, the percentage of larvae with tumors was calculated by dividing the number of larvae with tumors by the total amount of larvae. The mean percentage of larvae with tumors was calculated, along with the standard deviation. The percentage of pupae eclosed was calculated to measure the viability. This was calculated by dividing the number of pupae eclosed (meaning they were viable and left their protective casing) by the total amount of pupae. The mean percentage of eclosed pupae was calculated, along with the standard deviation.

Statistical analysis was completed using a chi-square test using the pooled data. To pool the data, in each of the 6 treatment groups, the amount of larvae with tumors were totaled and the amount of larvae without tumors were totaled. The viability data was also pooled in the same fashion. The chi-square test was set up with two variables: tumor/non-tumor and the RNAi

Group (Dome, Ebony, and White). Chi-square was used pairwise, meaning Dome compared with Ebony, Dome compared with White, and Ebony compared with White.

Results

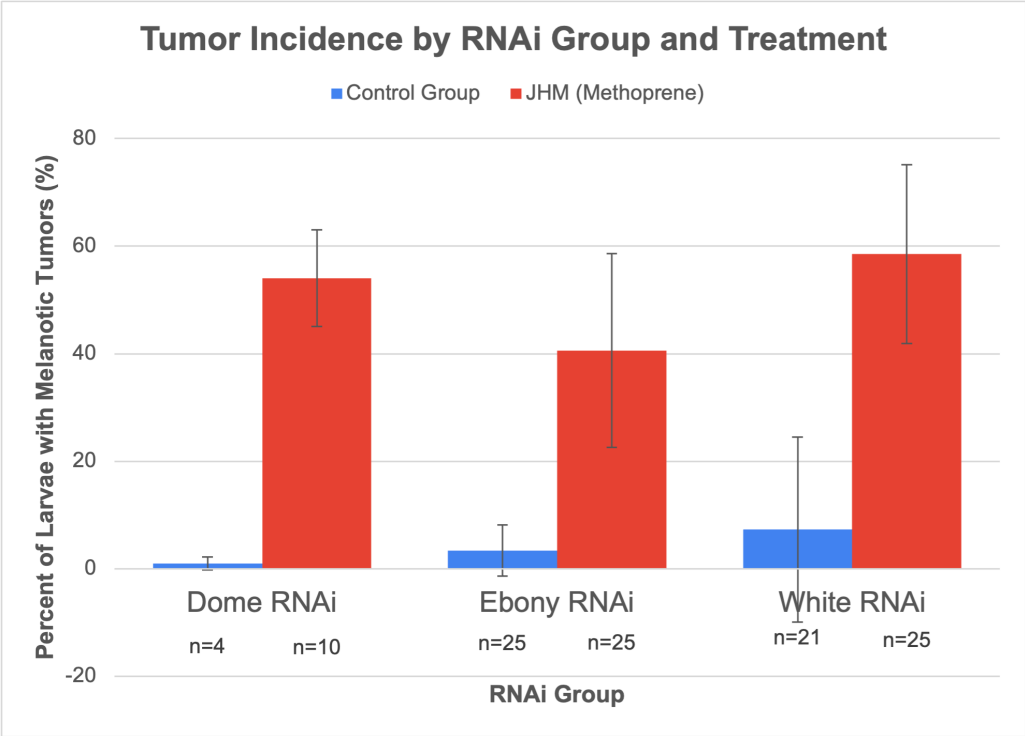


Figure 1. Juvenile hormone mimic (Methoprene) induces the formation of melanotic tumors regardless of gene knockdown:
Percent of Drosophila melanogaster larvae with melanotic tumors across 6 total experimental groups (3 different RNAi lines with either methoprene or ethanol treatment). Each bar represents the mean percentage of larvae with melanotic tumors, and each error bar represents one standard deviation. Sample sizes (# of vials) are indicated below the bars. Percentage of larvae was calculated by dividing tumors over total amount of larvae (tumors + non-tumors).

Juvenile hormone mimic (Methoprene) induces melanotic tumors in all RNAi groups. In Dome RNAi, 54% of methoprene-treated larvae had tumors, with a standard deviation of 8.99 (n=10), and 1% of ethanol-treated control larvae had tumors, with a standard deviation of 1.2 (n=4). In Ebony RNAi, 40% of methoprene-treated larvae had tumors, with a standard deviation of 18.0 (n=25), and 3.3% of ethanol-treated control larvae had tumors, with a standard deviation of 4.7 (n=25). In White RNAi, 58% of methoprene-treated larvae had tumors, with a standard deviation of 16.6 (n=25), and 7.3% of ethanol-treated control larvae had tumors, with a standard deviation of 17.2 (n=25). In the chi-square tests, Dome RNAi and Ebony RNAi had a significant difference, with a chi-square value of 46.1 and a p-value of <0.0001; Dome RNAi and White RNAi didn't have a significant difference, with a chi-square value of 2.72 and a p-value of 0.10;

and Ebony RNAi and White RNAi had a significant difference, with a chi-square value of 120.0 and a p-value of <0.0001.

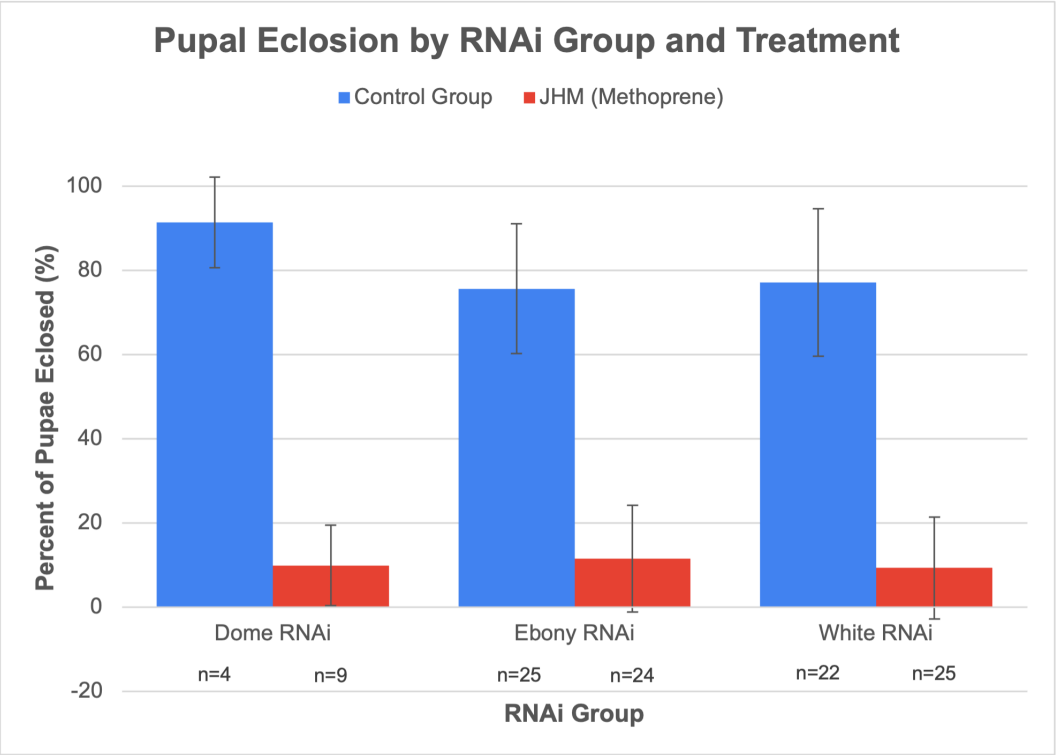


Figure 2. Mean percentage of pupal eclosion decreased drastically regardless of gene knockdown:

Percent of Drosophila melanogaster pupae eclosed (fly emerged from its pupal case) across 6 total experimental groups (3 different RNAi lines with either methoprene or ethanol treatment). Each bar represents the mean percentage of eclosed pupae, and each error bar represents one standard deviation. Sample sizes (# of vials) are indicated below the bars. Percentage of pupae eclosed was calculated by dividing eclosed over total amount of pupae (eclosed + full).

Juvenile hormone mimic (Methoprene) drastically decreases the percentage of pupal eclosion in all RNAi groups. In Dome RNAi, 9.9% of methoprene-treated larvae were empty and eclosed, with a standard deviation of 9.5 (n=10), and 91% of ethanol-treated control larvae were empty and eclosed, with a standard deviation of 10.7 (n=4). In Ebony RNAi, 11.5% of methoprene-treated larvae were empty and eclosed, with a standard deviation of 12.7 (n=25), and 75.6% of ethanol-treated control larvae were empty and eclosed, with a standard deviation of 15.4 (n=25). In White RNAi, 9.3% of methoprene-treated larvae were empty and eclosed, with a standard deviation of 16.6 (n=25), and 77.1% of ethanol-treated control larvae were empty and eclosed, with a standard deviation of 17.4 (n=25).

Discussion

After methoprene treatment, the percent of larvae with melanotic tumors of the three RNAi groups increased to 54% Dome RNAi, 40% Ebony RNAi, 58% White RNAi. All control groups that are not treated with methoprene show near 0% of tumor incidence, which demonstrates that the exposure of JHM leads to the formation of melanotic tumors regardless of the gene knockdown. Since Dome is the only transmembrane receptor on the JAK/STAT pathway, the similarity of tumor rates in three RNAi groups suggest that JHM-induced lamellocyte proliferation occurs independently of the Dome receptor.

Furthermore, the treatment of JHM significantly decreases the pupal eclosion rate among all control RNAi groups. (Dome RNAi shifts from 91% to 9.9%, Ebony RNAi decreases from 75.6% to 11.5%, White RNAi decreases from 77.1% to 9.3%). The exposure of JHM negatively disrupted pupal eclosion rates across all groups, causing a large proportion of Drosophila to die in the pupae and fail to eclose. In particular, the effects of JHM consistently remain similar across three groups of RNAi, which suggests that the knocked-down gene is not the primary reason that drives development failure. Therefore, the knockdown of Dome will not contribute either to increases or decreases to the amount of development failure caused by JHM exposure.

Our chi-square pairwise reveals that there is a significant difference in tumor formation between Dome RNAi vs. Ebony RNAi and a significant difference between White RNAi vs. Ebony RNAi. The significant difference is due to a lower percentage of tumors in Ebony RNAi. Further, our controls (Ebony RNAi and White RNAi) were significantly different, which shows that one of the controls had a notable influence in tumor formation. We hypothesized that Ebony RNAi was an inaccurate control because it directly increases melanin formation, which is what gives the tumors their dark color. We speculated that because Ebony RNAi gives the fly a darker pigment, it will be more difficult to identify the dark, melanized tumors, leading to many false negatives and thus a lower percentage of tumors. We also speculated that because a darker pigment necessitates more melanin, there is less melanin available to give the tumors a melanized and identifiable color. Therefore, we chose to compare Dome RNAi with White RNAi, which resulted in no significant difference. Without a significant difference, our hypothesis is not supported, and it is likely that JAK-STAT is not the pathway used to form melanotic tumors.

Using methoprene as the only JHM inducer may limit the generalizability of our experiment, as methoprene has its own specific chemical properties that may contribute to imbalances in Drosophila metabolism. To make the experiment more reliable in testing the receptor's impact on lamellocyte proliferation, multiple JHM inducers would be needed to measure the effect of the JAK/STAT pathway. In addition, the Dome RNAi control group included only 4 vials, while the White and Ebony control groups each contained 25 vials. The lack of samples in the Dome group may significantly affect the accuracy of our measurements and limit the reliability of the statistics; the result for Dome RNAi may not fully demonstrate that the knockdown of this gene does not affect melanotic tumor formation. Finally, the chi-square

pairwise tests showed significant differences in every comparison involving the Ebony group, which indicates variability among our controls and suggests that Ebony may not be the most reliable negative control to support our conclusion.

Moreover, because the Dome receptor doesn't directly influence tumor formation, JHM most likely acts through a different immune system regulation pathway, independent of the existing pathway. Future studies can explore hormone-immune system interactions and how chemical exposure affects immune responses.

An implication of this experiment is the elimination of JAK-STAT as a possible immune pathway utilized in the formation of melanotic tumors. This finding leads to future experiments that can test other immune pathways, such as Toll, to identify the specific immune pathway that mediates tumor formation. Because methoprene is widely used as an insecticide, when the immune pathway is identified, scientists can identify the specific side effects on mosquitoes, pests, and other species affected by the insecticide. Furthermore, because many immune pathways are conserved among species, identifying the immune pathway can also identify the side effects on the entire ecosystem.

In conclusion, our hypothesis that melanotic tumor formation will decrease when Dome RNAi is knocked down was not supported. Future directions for this experiment would be to increase the Dome RNAi sample size to reduce variability, use different JHMs to study their different effects, and to utilize a better control group that is not related at all to immune pathways. Other future experiments would be to test other immune pathways, such as Toll pathway, to identify the specific immune pathway that mediates melanotic tumor formation due to JHMs.

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